PDS - IMP PYQ Analysis

PYTHON FOR D.S. | BE05000231 | Sem 5 | GTU

New Syllabus 2024-25 | Old Papers Analyzed: 10 (W20-25)

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Chapter Reference

New Ch.	Covers
Ch. 1	Intro, Data Types, Operators, Decision Making, Loops, Functions
Ch. 2	File Handling, Exception Handling, Python-MySQL *(Zero PYQ, Blind Spot)*
Ch. 3	Data Science & Descriptive Statistics
Ch. 4	Exploring Python Libraries (SciPy, NumPy, Pandas, Scikit-learn, BeautifulSoup)
Ch. 5	Probabilistic & Inferential Statistics *(Zero PYQ, Blind Spot)*
Ch. 6	Data Preparation (cleaning, missing data, wrangling)
Ch. 7	Data Visualization (Matplotlib, Seaborn)

M.IMP Questions, High → Low Priority

> **Priority Rule:** Freq ≥ 3 OR appeared in ≥2 of last 3 exams OR 7-mark type asked ≥2x → **M.IMP**

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1. Missing Data Handling, Techniques, Imputation, Encoding Missingness

`(Ch. 6)` | **Freq: 6x** | **Max Marks: 7** | `Theory: 5` `Def: 1`

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2. Data Science Pipeline / Life Cycle, Stages & Phases

`(Ch. 3)` | **Freq: 5x** | **Max Marks: 7** | `Theory: 5`

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3. Pandas DataFrame Ops, shape/tail/describe, Read CSV, Slicing Rows & Cols

`(Ch. 6)` | **Freq: 5x** | **Max Marks: 7** | `Theory: 2` `Code: 3`

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4. NumPy vs Pandas, Comparison & Characteristics

`(Ch. 4)` | **Freq: 5x** | **Max Marks: 4** | `Theory: 5`

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5. Covariance & Correlation, Definition + Role in EDA

`(Ch. 3)` | **Freq: 5x** | **Max Marks: 4** | `Theory: 4` `Def: 1`

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6. Matplotlib Plot Types, Pie / Bar / Histogram / Boxplot (Code + Theory)

`(Ch. 7)` | **Freq: 8x combined** | **Max Marks: 7** | `Code: 5` `Theory: 3`

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7. Labels, Annotations & Legends in Matplotlib

`(Ch. 7)` | **Freq: 5x** | **Max Marks: 4** | `Theory: 4` `Code: 1`

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8. EDA, Definition + Tasks Involved

`(Ch. ~5/6, unmapped)` | **Freq: 5x** | **Max Marks: 7** | `Theory: 5`

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9. Scikit-learn, Classification & Regression Problems

`(Ch. 4)` | **Freq: 4x** | **Max Marks: 7** | `Theory: 4`

10. Python Data Structures, List vs Tuple vs Dict Comparison

`(Ch. 1)` | **Freq: 6x** | **Max Marks: 4** | `Theory: 6`

11. Categorical Variables, Definition & Handling

`(Ch. 6)` | **Freq: 4x** | **Max Marks: 4** | `Theory: 3` `Def: 1`

12. Sampling, Definition, Types, Techniques

`(Ch. 5)` | **Freq: 3x** | **Max Marks: 3** `Theory: 3`

13. Skewness & Kurtosis

`(Ch. 3)` | **Freq: 3x** | **Max Marks: 7** | `Code: 2` `Theory: 1`

14. Data Wrangling, Definition & Steps

`(Ch. 6)` | **Freq: 3x** | **Max Marks: 3** | `Theory: 3`

15. Z-score Standardization

`(Ch. 6)` | **Freq: 3x** | **Max Marks: 3** | `Theory: 3`

16. Removing Duplicates / Slicing-Dicing-Filtering Concept

`(Ch. 6)` | **Freq: 3x** | **Max Marks: 7** | `Theory: 2` `Code: 1`

IMP Questions, High → Low Priority

> **Priority Rule:** Freq = 2 OR 1 with **7 marks**, OR **3/4-mark** type asked ≥2x → **IMP**

1. String Slicing, Name Extraction Example

`(Ch. 1)` | **Freq: 3x** | **Max Marks: 4** | `Code: 3`

2. Fibonacci Series / Factorial Program, Recursion or Function

`(Ch. 1)` | **Freq: 4x** | **Max Marks: 7** | `Code: 4`

3. range() Function

`(Ch. 1)` | **Freq: 3x** | **Max Marks: 3** | `Theory: 3`

4. Reading Data from Text/CSV Files via Pandas

`(Ch. 2/6)` | **Freq: 4x** | **Max Marks: 4** | `Code: 4`

5. GroupBy Function in Pandas

`(Ch. 6)` | **Freq: 2x** | **Max Marks: 3** | `Theory: 2`

6. Sorting & Shuffling DataFrame

`(Ch. 6)` | **Freq: 2x** | **Max Marks: 4** | `Code: 2`

7. Importance of Data Visualization

`(Ch. 7)` | **Freq: 2x** | **Max Marks: 3** | `Theory: 2`

8. Supervised vs Unsupervised Learning

`(Ch. 4)` | **Freq: 2x** | **Max Marks: 3** | `Theory: 2`

9. Advantages / Role of Python in Data Science

`(Ch. 1/3)` | **Freq: 3x** | **Max Marks: 3** | `Theory: 3`

10. Scatter Plot with Trendline

`(Ch. 7)` | **Freq: 2x** | **Max Marks: 4** | `Theory: 1` `Code: 1`

📌 New Syllabus Blind Spots, High → Low Priority

> **Zero PYQ Coverage, Study Anyway.** Predictions only, not PYQ-backed.

1. Hypothesis Testing (One-Sample Test)

📍 `Ch. 5` | Likely angle: Definition + steps/example | `Theory, 4 marks`

2. Central Limit Theorem

📍 `Ch. 5` | Likely angle: Statement + significance in sampling | `Theory, 3 marks`

3. Probability Distributions, Binomial, Poisson, Normal, Exponential

📍 `Ch. 5` | Likely angle: Define + real-world example or PMF/PDF numerical | `Theory/Numerical, 7 marks`

4. Exception Handling in Python

📍 `Ch. 2` | Likely angle: try/except/finally code snippet with custom exception | `Code, 4/7 marks`

5. File Handling, Binary Files

📍 `Ch. 2` | Likely angle: Read/write binary file code example | `Code, 4 marks`

6. Python-MySQL Basic Operations

📍 `Ch. 2` | Likely angle: Connect, cursor, execute query code | `Code, 7 marks`

7. SciPy stats Package

📍 `Ch. 4` | Likely angle: Define + one function usage example | `Theory, 3 marks`

8. Seaborn Basics

📍 `Ch. 7` | Likely angle: One plot type, compared vs Matplotlib | `Theory/Code, 3/4 marks`

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9. Point vs Interval Estimation

📍 `Ch. 5` | Likely angle: Differentiate with example | `Theory, 3 marks`

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10. PMF / PDF / CDF Definitions

📍 `Ch. 5` | Likely angle: Define each, differentiate | `Theory, 4 marks`

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11. Bitwise / Membership / Identity Operators

📍 `Ch. 1` | Likely angle: Code output prediction | `Code, 3 marks`

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12. Data Measurement Scale, Nominal / Ordinal / Interval / Ratio

📍 `Ch. 3` | Likely angle: Define + example per scale | `Theory, 4 marks`

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